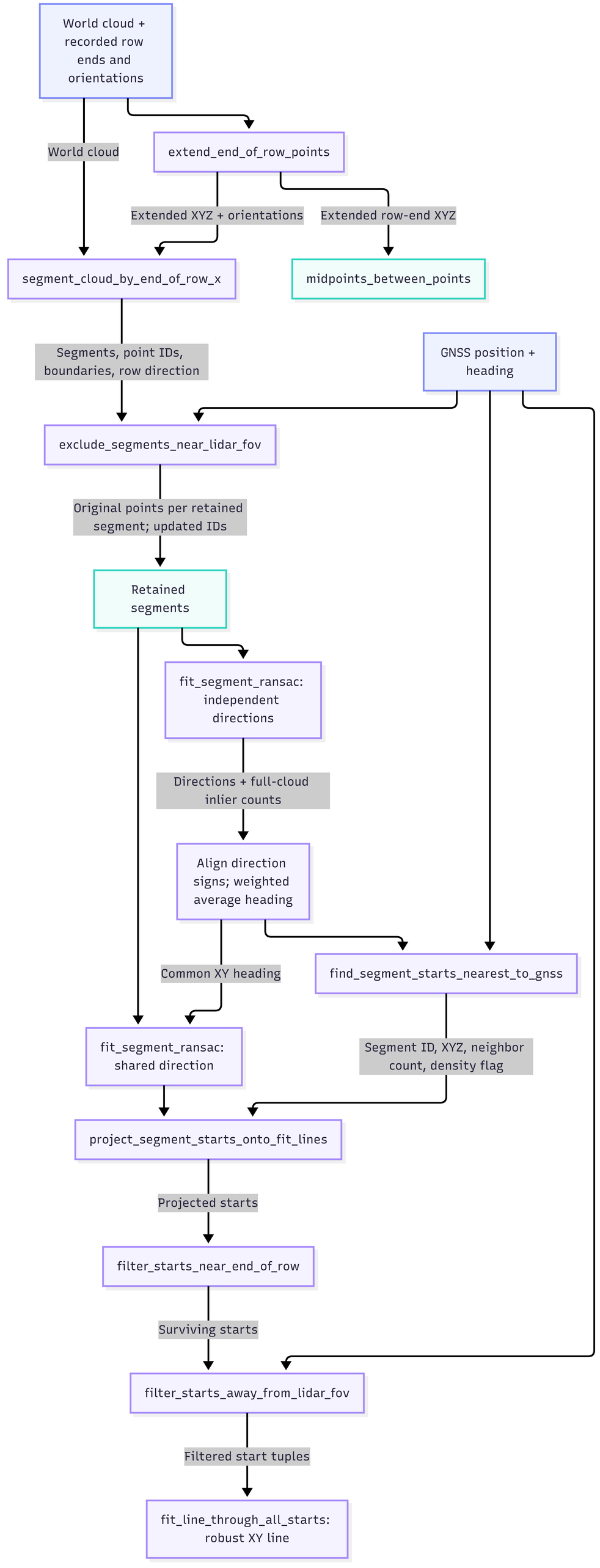

World cloud + recorded row ends and orientations
World cloud
extend_end_of_row_points
Extended XYZ + orientations
Extended row-end XYZ
segment_cloud_by_end_of_row_x
midpoints_between_points
Segments, point IDs, boundaries, row direction
GNSS position + heading
exclude_segments_near_lidar_fov
Original points per retained segment; updated IDs
Retained segments
fit_segment_ransac: independent directions
Directions + full-cloud inlier counts
Align direction signs; weighted average heading
Common XY heading
find_segment_starts_nearest_to_gnss
fit_segment_ransac: shared direction
Segment ID, XYZ, neighbor count, density flag
project_segment_starts_onto_fit_lines
Projected starts
filter_starts_near_end_of_row
Surviving starts
filter_starts_away_from_lidar_fov
Filtered start tuples
fit_line_through_all_starts: robust XY line